FOC-MC-R1

Sensored Field-Oriented-Control Motor Controller for a Quasi-Direct-Drive Robot Joint

Technical Reference Manual · Rev 1 hardware · Firmware 0.6.1 (d321545) · Document 1.0, 2026-09-30 · PRELIMINARY

This manual consolidates the full project record: requirements, design decisions and their history, circuit design, simulation, thermal analysis, PCB and fabrication, firmware, GUI, bring-up results and known issues. Values are marked **design** (calculated or specified), **sim** (SPICE) or **measured** (Rev 1 bench, 12 V supply). Where earlier project documents disagree with this one, Section 19 lists the correction.

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1 Overview

1.1 Features

- Three-phase inverter, 6 × Infineon ISC030N10NM6 (100 V, 3.0 mΩ max, Q_g 55 nC, PG-TDSON-8 5×6 mm)
- Bus 36–56 V (12S Li-ion); 10 A RMS continuous phase current (design)
- STM32G431RBT6 (Cortex-M4F, 170 MHz); TI DRV8323S gate driver: SPI IDRIVE/TDRIVE, charge pump, 3 CSAs, VDS OCP
- Three 3 mΩ low-side Kelvin shunts; CSA gain 20; centre-aligned synchronous sampling
- AS5047P 14-bit absolute magnetic encoder at board centre; auxiliary SPI encoder connector
- 25 kHz centre-aligned PWM; control ISR every 40 µs; 200 ns dead time
- CAN-FD (FOCCAN/1) to host; UART (FOCLINK/1) to PC GUI; SWD
- Build-time power-stage lock; tiered approved limits; hardware break on driver fault; input-staleness supervision
- On-board motor identification: phase check, R, L_d/L_q, flux linkage; on-MCU plant simulation
- 70.1 × 70.1 mm, 6 layers, 4 × M3

1.2 Status

Area	State (2026-09-30)
Hardware	Rev 1 fabricated and assembled by JLCPCB. Powered on 12 V bench supply only.
Power stage	Switching validated: 200 ns dead time, zero overlap, duty ±0.3 %. Switch node not yet scoped.
Motor control	Motor ID and open-loop I/f to 2000 rpm on bench motor. Closed-loop FOC not run (encoder magnet not fitted).
Comms	FOCLINK/1 and GUI in daily use. CAN-FD implemented, not run on a bus.
Not validated	Operation above 12 V, any mechanical load, closed-loop torque/velocity/position, thermal at rated current.

2 Requirements

Application: one joint of a cat-scale bipedal robot using quasi-direct drive (QDD), a gear ratio of roughly 6:1 to 10:1 so the joint is backdrivable and torque tracks motor current almost directly. Target motor: ODrive M8325s (20 pole pairs). Bench motor: T-Motor MN601-S KV320 (14 pole pairs).

Requirement	Value	Derived from
Bus	12S Li-ion: 36 V cutoff, 44.4 V nominal, 50.4 V full; 56 V stated max; 60 V ceiling	Battery
Continuous phase current	10 A RMS	Joint torque budget
Current-sense accuracy	Primary spec: sense error is torque error in QDD	QDD architecture
Position feedback	Absolute, available at zero speed	Joint must hold torque at standstill
Switching frequency	Inaudible (> 20 kHz). 40 kHz planned, 25 kHz implemented	Audibility; loss; ISR budget
Nodes	6 boards on one bus, CAN-FD, hardware-distinguishable	Robot topology
Input protection	Hot-swap and inrush handled once, on an upstream PDB	Bus-level function

Requirement	Value	Derived from
Mass/area	Minimise: board hangs on a limb; ≤ 80 × 80 mm	Mechanical
Safety	Fault = coast; no uncommanded torque; independent stop path	Legged robot

Note on f_{sw}: the real requirement is audibility. The pole-pair check (20 pp × 3500 rpm ≈ 1.17 kHz electrical, ×20 samples ≈ 23 kHz) is a sanity check added afterwards, not the origin of the number.

3 Design History

The design followed an underdamped path: undersized, then overcomplicated, then trimmed to the requirement.

Rev	Configuration	Outcome / lesson
A	STM32F405; CSD18540Q5B 40 V MOSFETs	40 V device on a 50.4 V bus: out of rating at rest. Rule adopted: check V_{DS} against worst-case transient first.
B	STM32G431CBU6; DRV8323S; IPT015N10N5 (1.5 mΩ)	Loss inversion: lowest $R_{DS(on)}$ had highest total loss at 40 kHz. Selection moved to figure of merit.
C	LM5069 hot-swap + SOA pass FETs, dual XT60, USB-C + CH340X, 3 regulators + TPS2116 mux, CAN ID ladder, off-board encoder, 19 sheets	Overcorrection: every fault protected at board level. Each addition locally justified; placement in the system never questioned.
1	Hot-swap to PDB; SWD replaces USB; LMR36520 + TLV75733 only; on-board encoder; TDSON-8 FETs; direct-solder power terminals	Strip-to-minimum pass: each block classified as required or discretionary, and assigned to the right board. Built and fabricated.

3.1 Parts removed and why

Removed	Why it existed	Why it went
LM5069 hot-swap front end	Inrush, UV/OV, fault current limit	Bus-level function; once on the PDB instead of ×6
USB-C + CH340X bridge	Console and config	SWD/RTT does the job; frees BOOT0/pin conflicts
Gate R + parallel diode network	Asymmetric turn-on/off	DRV8323 IDRIVEP/IDRIVEN provides it internally
LED buffer MOSFETs	5 V / 3.3 V mismatch	Mismatch gone once LEDs moved to 3V3; 3.9 mA per LED
Separate analog LDO (NCP176) + mux	Clean ADC reference	Removed in trim; analog now shares 3V3 (E-08)
CAN node-ID resistor ladder	16 IDs on one ADC pin	Deferred to Rev 2
Off-board-only encoder	Mounting freedom	Unshielded SPI through a moving joint was the top field-failure risk

4 Ratings and Operating Conditions

4.1 Absolute maximum ratings

Parameter	Min	Max	Unit	Limited by
Bus voltage, continuous	—	51	V	SMCJ51CA stand-off (V_{WM}); breakdown band begins 56.7 V
Bus voltage, transient	—	65	V	DRV8323 V_M and LMR36520 V_{IN} abs max
MOSFET V_{DS} / V_{GS}	−20	100 / 20	V	ISC030N10NM6
Junction temperature	−55	175	°C	ISC030N10NM6
Logic I/O	−0.3	3.6	V	STM32G431 (non-FT pins)

The TVS clamps at 82.4 V (at I_{PP} 18.2 A), above both 65 V ratings; there is no on-board protection between 65 V and the clamp (E-01).

4.2 Recommended and validated operating conditions

Parameter	Min	Typ	Max	Unit	Validated on Rev 1
Bus voltage	36	44.4	51	V	12 V only
Phase current, continuous	—	—	10	A RMS	3 A; OC trip set 4.5 A
Phase current, CSA linear range	−23.3	—	23.3	A peak	Not exercised
Ambient	0	25	40	°C	Bench
PWM frequency	—	25	—	kHz	Yes
CAN bit rate	—	1	—	Mbit/s	Not run

5 Electrical Characteristics

Parameter	Value	Type	Notes
$R_{DS(on)}$ 25 °C / hot	2.6 / ~6 mΩ	design	~2× at 125–150 °C
Dead time	200 ns, zero overlap	measured	Edges timed against TIM1->CNT
PWM duty accuracy	±0.3 %	measured	
IDRIVEP / IDRIVEN	0.33 / 0.66 A	config	See 8.2 before changing
Charge-pump demand	4.1 mA of 25 mA	design	$3 \cdot Q_g \cdot f$ at 25 kHz; 6.1× margin
Peak V_{DS}, 50.4 V, 14.1 A	52.6 V (LS) / 51.4 V (HS)	sim	Loop 2.4 nH, IDRIVE 0.33/0.66
E_{on} / E_{off} at 14.1 A	18.4 / 2.7 µJ	sim	E_{on} includes body-diode Q_{rr}
Current sense ideal FS / linear range	±27.5 / ±23.3 A	design	Gain 20; linear range set by CSA output swing
Current resolution	13.4 mA/LSB	design	12-bit, 3.3 V
CSA offsets A/B/C	1649 / 1632 / 1622 mV ±1 mV	measured	Normal-mode cal, bridge idle
Current noise	11–14 mA rms	measured	Per sample
Bus-sense ratio / resolution	0.05089 / 15.8 mV/LSB	design	24.9 k + 12.4 k over 2 k
Bus-sense bandwidth	838 Hz	design	1.9 kΩ source, 100 nF; monitor only
Control ISR period / cost	40 µs / 18 idle, 29 spin	measured	+~1 µs per scope signal
DRV8323 SPI clock	332 kHz	measured	Rev 1 limit (E-04)
Bus-divider quiescent draw	1.28 mA at 50.4 V	design	Unswitched

6 System Architecture

```
H8(+) -> SMCJ51CA -> +VPROT
  +VPROT -> 3x100uF polymer + 9x4.7uF MLCC -> bridge 6x ISC030N10NM6 -> H1/H2/H3 (A/B/C)
  +VPROT -> DRV8323S (VM, VDRAIN, charge pump)
  +VPROT -> LMR36520 (400 kHz) -> +5V -> TLV75733 -> +3V3 (digital + analog)
Shunts 3 mOhm -> DRV8323 CSA x3 -> ADC2 (A) / ADC1 (B, C, VBUS), injected, TIM1_TRGO2
AS5047P -> SPI1 ; DRV8323 <- SPI3, 6x PWM (TIM1), ENABLE ; nFAULT -> TIM1_BKIN
Host <- CAN-FD J4/J5 -> TCAN1044A -> FDCAN1 ; PC <- UART J1 -> USART1 + DMA2
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6.1 Ground domains

- PGND (FET sources, shunt returns, DRV8323 PGND pin 40), AGND (DRV8323 AGND pin 32, CSA filters, dividers), DGND: joined at one star point at the shunts.
- CHASSIS isolated: one DC bond (R58, 0 Ω at H6) plus 1 nF / 1 kV at every mounting hole. A second DC bond would form a loop through cable shields.
- L2 and L5 are solid, unbroken ground planes over the whole board, including under the power stage.

6.2 Connectors

Ref	Type	Position (mm)	Function / pinout
H3 / H2 / H1	3.2 mm PTH solder	18.5 / 30.0 / 41.5, y 71	Phase C / B / A
H8 / H9	3.2 mm PTH solder	54.0 / 63.0, y 71	V_{BUS}+ / GND; 18 AWG, ≤ 200 mm, twisted
J1 / J3	JST-GH 4	27.00 / 17.25, 9.50	UART (FOCLINK/1) / I2C3
J4 / J5	JST-GH 4	38.00 / 44.75, 11.00	CAN-FD in / out
J2 (DNP)	Header 3	33.25, 9.96	CAN breakout, along +y: GND, CANH, CANL
J6	JST-GH 7	60.75, 9.50	AUX encoder SPI: p4 CS, p5 CLK, p6 MISO, p7 MOSI (100 Ω series); p1–3 supply/GND
J10 / J11	JST-GH 2	70.50, 26.25 / 19.00	Motor NTC / E-STOP (non-functional, E-02)
J7 (DNP)	Header 5, 2.54	7.75, 32.50	SWD along +y: GND, SWDIO, SWCLK, NRST, 3V3
SW1	DPDT slide	9.75, 17.25	Beside CAN split termination R45/R46 60.4 Ω + C68 47 nF; function to confirm

6.3 MCU pin assignment (selected)

Signal	Pin	Peripheral	Note
PWM A/B/C high	PA8/PA9/PA10	TIM1 CH1–3	PA9/PA10 high enable UCPD pulldowns on PB6/PB4 (F-01)
DRV_ENABLE	PC7	GPIO	Early pin map used PC6
DRV_nFAULT	PA15	TIM1_BKIN	Hardware break; AOE off
E-STOP	PC3	TIM1_BKIN2	Follows DRV_ENABLE via R41 (E-02)
Encoder CS	PA4	SPI1	Early pin map used PC4
SOA / SOB / SOC	PB11 / PB12 / PB14	ADC12 / ADC1 / ADC1	B and C never sampled together
UART1_TX / LED	PB6 / PB4	USART1 / GPIO	UCPD dead-battery pulldowns

Signal	Pin	Peripheral	Note
FDCAN_RX	PB8	FDCAN1	Shared with BOOT0 (E-05)

7 Control Theory

7.1 Field-oriented control

Six-step commutation gives ~13 % torque ripple and no control of current angle. FOC transforms phase currents into a rotor-synchronous frame where they are DC and PI control applies directly.

```
Clarke: (ia, ib, ic) -> (i_alpha, i_beta)          Park: (i_alpha, i_beta) + theta -> (i_d, i_q)
i_d = 0 (surface-PM: no useless flux current);   i_q -> torque:  T = 1.5 x p x lambda x i_q = Kt x i_q
Inverse Park -> SVM: ~15 % more bus utilisation than sinusoidal PWM
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- Park needs rotor angle θ. At zero speed back-EMF is zero, so sensorless observers have no signal; a joint must hold torque at standstill, hence an absolute encoder.
- Current is sampled at the PWM centre where all low-side FETs conduct; the third current is reconstructed from $i_a+i_b+i_c = 0$.

7.2 Open-loop I/f (used for bring-up without encoder)

- Regulate current magnitude, rotate its angle at a commanded rate; the rotor follows. Chosen over V/f because with R = 98 mΩ a raw voltage command is ~1 A per 1 % duty at 12 V.
- Minimum current is set by breakaway torque (≈ 37 mN·m), not speed: 0.5 A (14.6 mN·m) fails, 1.5 A spins.
- I/f has almost no electrical damping. The rotor swings at $f \approx \sqrt{(K_t \cdot I \cdot p/J)}/2\pi$ (~14 Hz at 1.5 A, ~20 Hz at 3 A).
- Uncompensated ~1.5-period control delay gives negative damping $\propto \omega_e$: hunting above ~300 rpm, pull-out at 530–700 rpm. Fix: rotate the output vector forward by $\sim 2 \cdot dt \cdot \omega_e$. Reproduced in tools/if_loop_model.py before flashing.
- Voltage limit is a speed limit: $rpm_{max} = K_V \cdot \sqrt{3} \cdot \sqrt{(V_{lim}^2 - (R \cdot I)^2)}$ (Fig. T2).
- Coast-safe speed ≈ $V_{bus} \cdot K_V$ (~3400 rpm at 12 V); firmware caps at 0.8× (2700 rpm). Above it a coasting rotor rectifies back-EMF into a supply that cannot sink it.

8 Power Stage Design

8.1 MOSFET selection

Metric: figure of merit $R_{DS(on)} \times Q_g$. At 40 A RMS, 40 kHz, 50 V, I_{DRIVE} 1 A:

$R_{DS(on)}$	P_{cond}	P_{sw}	P_{total}
1.50 mΩ (IPT015N10N5)	1.80 W	3.85 W	5.65 W (worst)
2.24 mΩ	2.69 W	1.58 W	4.27 W
2.70 mΩ	3.24 W	1.36 W	4.60 W

ISC030N10NM6	Value
$V_{(BR)DSS}$; $V_{GS(th)}$ min/typ/max	100 V; 2.3 / 2.8 / 3.3 V
$R_{DS(on)}$ typ / max, V_{GS} 10 V	2.6 / 3.0 mΩ
Q_g / Q_{gs} / Q_{gd} (max) / $V_{plateau}$	55 / 18 / 9.1 (14) nC / 4.6 V
C_{iss} / C_{oss} / C_{rss} at 50 V	4000 / 900 / 15 pF
Q_{oss}; Q_{rr} at 100 / 1000 A/µs	101 nC; 56 / 266 nC
R_G; R_{thJC} bottom; R_{thJA} (6 cm²)	1.05 Ω; 0.36 typ / 0.72 max °C/W; 50 °C/W

JLCPCB's 100 V TOLL stock targets 200–360 A BMS use (Q_g 165–252 nC), so the selected part is an extended/consigned line. TOLL was replaced by TDSON-8: six TOLL footprints (~700 mm²) were over-provisioned for < 1 W per FET at 10 A.

8.2 Gate drive

- DRV8323 IDRIVE sets slew: $I_{DRIVE} \approx Q_{gd}/t_r$. Series gate resistors are 0 Ω, so IDRIVE is the only slew control.
- Charge pump supplies high-side gates only: $I_{VCP} \geq 3 \cdot Q_g \cdot f_{PWM}$ = 4.1 mA at 25 kHz (8.3 mA at 40 kHz, $Q_{g,max}$ 69 nC) vs 25 mA.
- TDRIVE (500–4000 ns): VGS-sensed adaptive dead time; ISTRONG 2 A pulldown on the opposite gate while its partner switches; IHOLD 10 mA source / 50 mA sink after TDRIVE; gate fault detection.
- **Do not raise IDRIVE without scoping.** Simulation (Fig. S3) shows high-side recovery overshoot rising from 51 V (0.33 A) to 69 V (0.57 A) and 111 V (1.0 A). Switching energy falls with IDRIVE, so the trade is loss against overshoot.
- Fault state is coast. Shorting all low sides at speed would drive ~180 A (19 V back-EMF into ~100 mΩ).

8.3 Snubbers

Drain-source RC footprints per leg (R23/C35 bus–phase, R24/C36 phase–source; 3R9 / 2n2) are fitted DNP. Earlier LTspice: ring 89.7 MHz, < 1 V, self-damped in ~40 ns; extracted L_{par} 2.44 nH, C_{par} 1271 pF, Z_0 1.39 Ω; 32 R/C combinations gave no benefit. If EMC ever requires them: R ≈ 1.4 Ω, C ≈ 3.8–6.4 nF. Dissipation if fitted ≈ 64–80 mW, not $C \cdot V^2 \cdot f$ (224 mW), because the edge is slow relative to RC.

8.4 DC link

- Sized by ripple-current capacity (~7.8 A RMS at 10 A/phase), not capacitance.
- Built: 3 × 100 µF/100 V polymer ≈ 7.5 A capacity (~96 % loaded); design froze 5 (12.5 A). MLCCs (9 × 4.7 µF/100 V, ~40 % at 50 V) close the HF loop and carry ~7–10 % of ripple.
- Bulk sits 20–45 mm from the far half-bridge; at 40 kHz, 40 nH ≈ 10 mΩ against ~35 mΩ ESR, so placement is acceptable. The HF loop is closed by MLCCs ~6 mm from the FETs.

9 Switching Simulation

9.1 Bench

- Tool: ngspice 42 (re-run 2026-09-30). Earlier LTspice leg-A results are quoted for comparison. Files: simulation/legA_dpt_bench.cir, ISC030N10NM6_fit.lib, sim_results.csv.
- Topology from MOSFETs.kicad_sch leg A: supply 20 mΩ + 100 nH → 300 µF bulk (12 mΩ) → 15 nH → 5.6 µF MLCC → HF loop 2.4 nH (package, shunt, ESL) → half-bridge; 3 mΩ shunt; 20 µH load across high side.
- Gate drive: DRV8323-style current sources (IDRIVEP/IDRIVEN), IHOLD 50 mA, ISTRONG 2 A for 500 ns at each opposite-side edge. Resistive-sink variant reproduces the earlier LTspice driver.
- Tests: double pulse (high side held off, body diode freewheels); complementary edge (high side conducts, off 200 ns before low-side turn-on).
- Numerical aids: 50 pF node capacitances and 5–50 Ω damping across parasitic inductors (represents plane capacitance and skin-effect loss).

Model validation	Datasheet	Fit (ngspice)	Earlier LTspice fit
$R_{DS(on)}$ at 10 V, 25–50 A	2.6 mΩ typ	2.55–2.57 mΩ	2.63–2.67 mΩ
Miller plateau at 25 A	4.6 V	4.57 V	fitted
Q_g 0–10 V	55 nC typ	53.8 nC	53.5 nC
Q_{gd}	9.1 nC typ	9.45 nC	9.1 nC

Model: VDMOS Vto 3.0, Kp 19.5, mtriode 4.8, Cgdmax 1.3 nF, Cgdmin 15 pF, Cgs 4.3 nF, Cjo 11.5 nF, tt 8 ns. Body-diode recovery is abrupt (no soft-recovery term), so recovery overshoot is pessimistic.

Fig. S1 Leg-A double pulse, 50.4 V / 14.1 A, IDRIVE 0.33/0.66 A, ISTRONG on

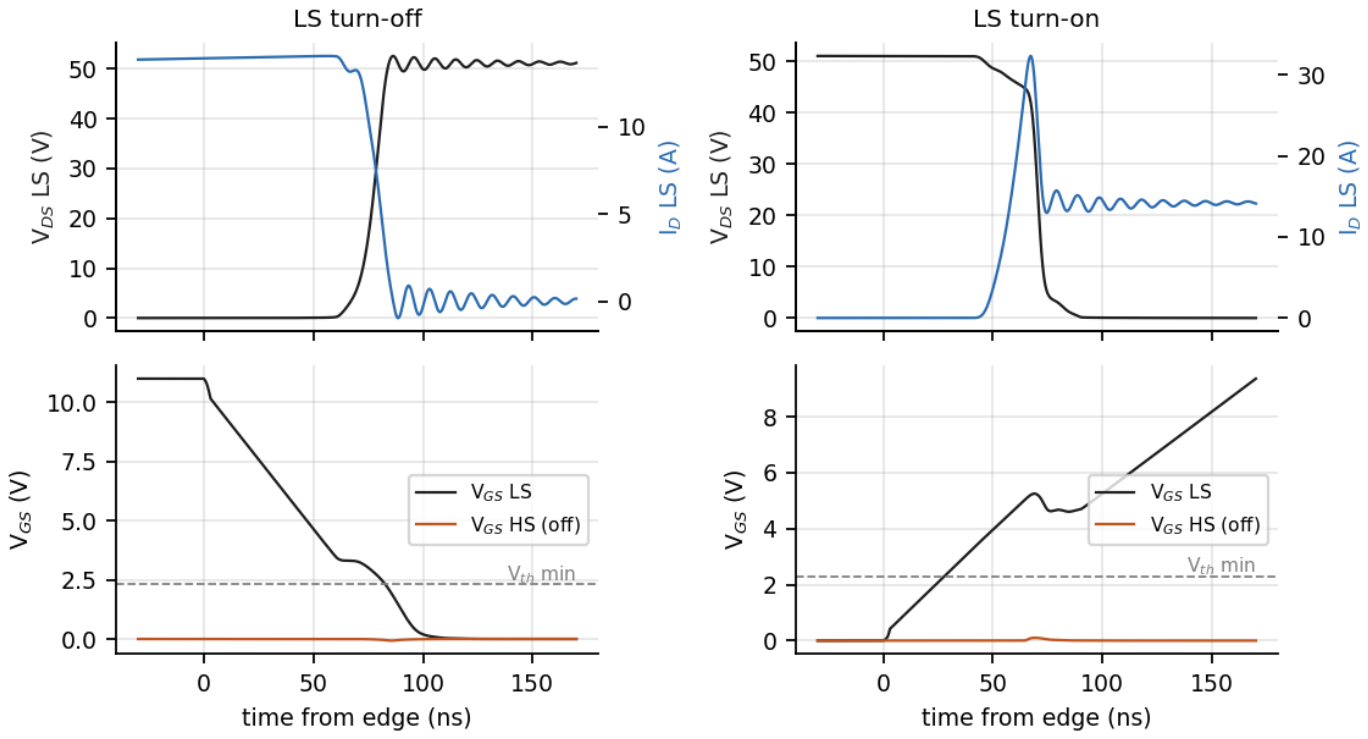


Fig. S1. Double pulse at 50.4 V / 14.1 A, IDRIVE 0.33/0.66 A, ISTRONG active. Turn-off: 52.6 V peak, ring self-damps. Turn-on: 32 A peak (recovery + C_{oss}) on a 14.1 A load; off-gate disturbance 0.10 V.

Fig. S2 Shoot-through margin depends on the gate-driver model

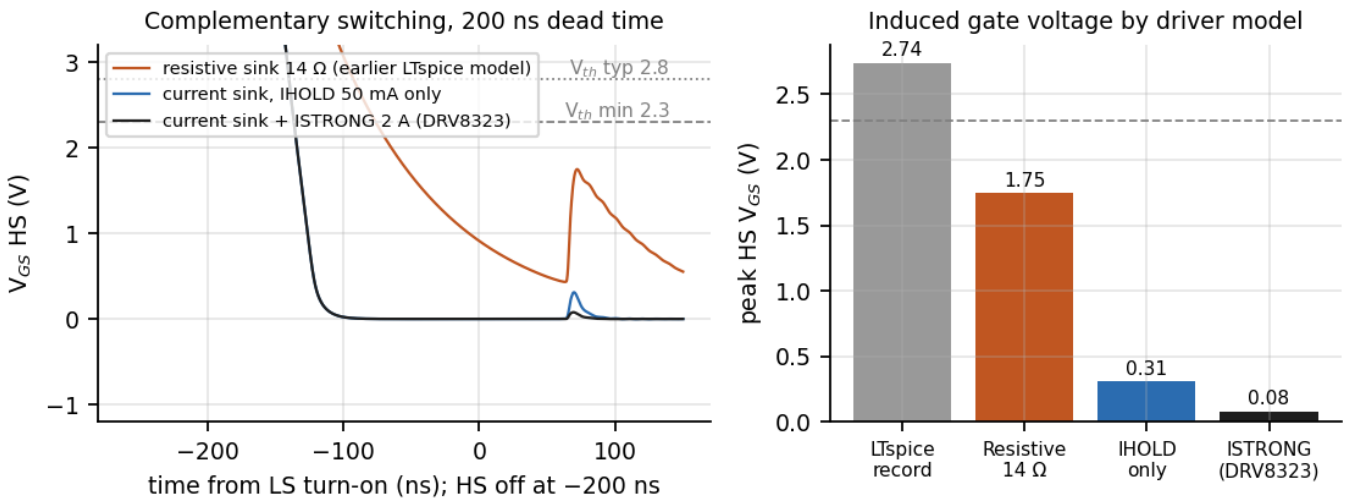


Fig. S2. Induced high-side gate voltage at low-side turn-on after 200 ns dead time. The earlier 0.06 V headroom result depends on a resistive-sink driver model; with the DRV8323 current sink and ISTRONG it falls to 0.08–0.31 V.

Fig. S3 IDRIVE sweep, 50.4 V, ISTRONG on (diode recovery model is abrupt: HS peak is pessimistic)

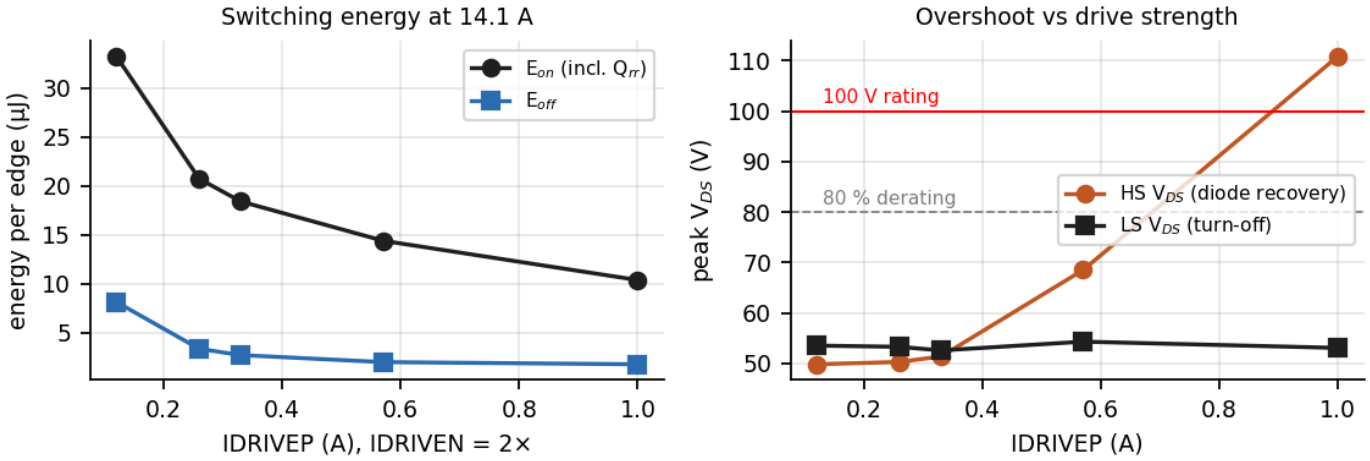


Fig. S3. IDRIVE sweep. Faster drive cuts E_{on} (33 → 10 µJ) but raises high-side recovery overshoot past the 100 V rating at 1.0/2.0 A. The earlier recommendation to raise IDRIVE is withdrawn pending measurement.

Fig. S4 Overshoot vs loop inductance (50.4 V, 14.1 A)

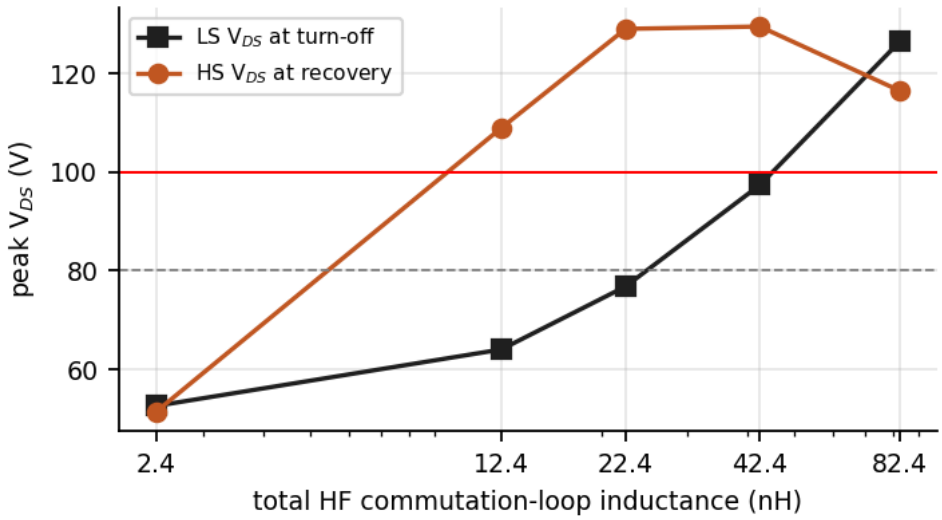


Fig. S4. Overshoot vs total HF commutation-loop inductance. Layout budget: keep the loop ≤ ~10 nH to stay within 80 % derating at 50.4 V. Fabricated layout is estimated at ~2.4 nH (earlier LTspice extraction 2.44 nH).

9.2 Findings

Result	Value (sim)	Consequence
Peak V_{DS}, as-built	52.6 V LS / 51.4 V HS	~48 % margin to 100 V
Induced HS V_{GS}, DRV8323 model	0.08 V (ISTRONG) / 0.31 V (IHOLD)	> 2 V margin to $V_{th,min}$; confirm by probe
Induced HS V_{GS}, resistive 14 Ω	1.75 V (LTspice record 2.74 V)	Explains the earlier finding
HS recovery overshoot vs IDRIVE	51 / 69 / 111 V at 0.33 / 0.57 / 1.0 A	Keep 0.33/0.66 until scoped
LS overshoot vs loop L	53 / 64 / 77 / 97 V at 2.4 / 12.4 / 22.4 / 42.4 nH	Loop area is the primary layout variable
E_{on} / E_{off} at 14.1 A	18.4 / 2.7 µJ	Feeds Section 10

10 Thermal

Fig. S5 MOSFET loss and junction temperature vs phase current (50.4 V, 25 kHz)

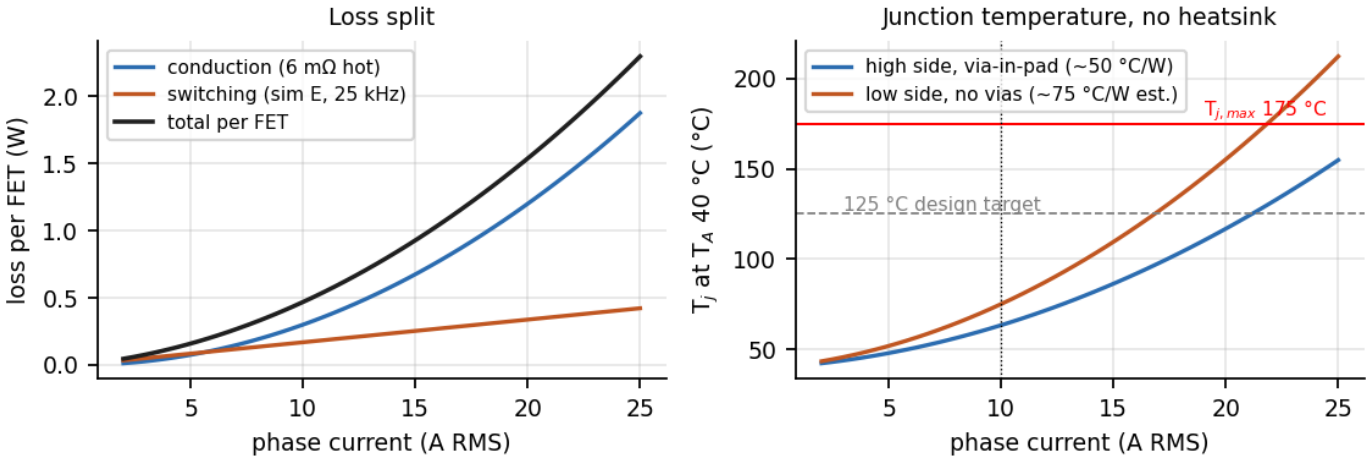


Fig. S5. Loss and junction temperature per FET vs phase current at 25 kHz, 50.4 V, T_A 40 °C, no heatsink. Switching loss from simulated E_{on}+E_{off} scaled with current and averaged over the sine (2/π).

Condition	Value	Source
T_j at 10 A RMS, HS / LS	~63 / ~73 °C	Fig. S5
Six-FET total, running, 25 / 40 kHz	3.9 / 5.3 W	Thermal model
Worst single FET, stall, 25 / 40 kHz	0.93 / 2.34 W	Stall is switching-dominated
Junction-to-sink path	~17.4 K/W	Via array ~13 K/W dominates
R_{thJC} bottom / top	0.72 / 20 K/W max	Heatsink goes under the board

Condition	Value	Source
Shunt dissipation, 10 / 20 A RMS	0.30 / 1.20 W	3 W part

- High-side FETs: 16 × 0.25 mm via-in-pad each into the inner +VPROT plane. Low-side FETs: none (switch node kept on L1 for EMI); ~80–100 mm² of top copper each, R_{thJA} estimated ~75 °C/W.
- Board NTC R37 sits at board centre (39.00, 40.25), 22 mm from the FETs, and under-reads; calibrate against a low-side case thermocouple.
- **Continuous rating is 10 A RMS** until low-side case temperature is measured. The board is sense-limited (±23.3 A) before it is thermally limited with a heatsink.

11 Sensing

11.1 Phase current

- Shunt and gain are one decision: 3 mΩ × 20 gives ±27.5 A ideal, ±23.3 A peak (16.3 A RMS) linear. Gain 40 → ±13.75 A (too low). Gain 10 → ~±46 A if thermals allow.
- Sample timing, not filtering, rejects switching noise. TIM1 CH4 (no pin) → OC4REF at counter peak → TRGO2 → injected ADC. RC 56 Ω + 2.2 nF (f_c 1.29 MHz, τ 123 ns) is edge damping; allow ≥ 10τ before sampling.
- Calibrate offsets in normal mode with the bridge idle, on every wake. DRV8323 CAL mode gives 0.15–0.46 A error on Rev 1.
- Signs proven on the motor by PHASE_CHECK: A +1, B +1, C −1. Datasheet-derived signs were wrong for all three.
- With the driver asleep the CSA outputs are meaningless (≈ −27 / −24 / +26 A after offset); the OC check runs only when awake and calibrated.
- Resistance fits are valid only where every phase current exceeds the PWM ripple; otherwise dead-time drop inflates R ~4×.

```
I_phase = sign x (V_ADC - V_offset) / (20 x 0.003)     1 LSB = 13.4 mA
V_BUS   = code x 3.3 / 4096 / 0.05089                   1 LSB = 15.8 mV
```

11.2 Bus voltage and temperature

- V_{BUS} divider 24.9 k + 12.4 k (split so no 0402 exceeds 50 V working voltage) over 2 k; 1.9 kΩ source with 100 nF: 838 Hz. Adequate for monitoring and regen fold-back, too slow for a transient trip.
- Board NTC (R37, 10 k), motor NTC (J10), MCU temperature and VDDA are read by the slow ADC sequence every 10 ms, one conversion per start (F-03).
- All slow readings carry an age; older than 500 ms refuses ARM and faults an armed board (fw 0.6.1).

12 Protection and Safety

Layer	Mechanism	Limit / behaviour
Input transient	SMCJ51CA TVS	Clamp 82.4 V (above 65 V ICs, E-01)
Input UV/OV, inrush	Upstream PDB	Not on this board
Shunt OCP (SEN_OCP)	DRV8323	Min SEN_LVL 0.25 V → 83 A; default 1.0 V → 333 A. Backstop only
VDS OCP	DRV8323	VDS_LVL 0.06 V → ~13 A hot / 20 A cold; spread ~2× over T_j
Firmware OC	Control ISR	Working limit; 4.5 A trip in current bench tier
Driver fault	nFAULT → TIM1_BKIN	Hardware output disable, independent of software
Output-lock invariant	ISR + main loop	Illegal MOE/AOE/CCxE bits cleared, FLT_SAFETY latched
ISR overrun	FLT_ISR	Outputs off within one period
Dead-man	Main loop	100 ms
Watchdog	IWDG	500 ms
Input staleness	Firmware	> 500 ms old: refuse ARM / fault
Regen	Regen guard	Controlled deceleration above ~120 rpm held; every stop is a coast
E-stop	J11 → PC3/BKIN2	Non-functional on Rev 1; supply switch is the E-stop

12.1 Safety process

- APP_PWM_OUTPUTS_ALLOWED = 0 in the repository, always. Bench builds set 1 locally; the staged file is grepped before every commit.
- Each new use of the bridge is a separately reviewed scope with limits in CLAUDE.md, enforced three times (Ctrl_Run, Ctrl_SetRef, step function).
- Raising a limit is a code change, staged, with the trip moved first (current cap 0.5 → 2 → 3 A).
- Parameters default 0 = UNSET; ARM/RUN refuse with a reason code until protections, R, L and gains are set.
- STOP in the GUI is a software stop and is never labelled E-STOP.

13 Firmware

13.1 Architecture

Layer	Modules
Entry	main.c (CubeMX init; USER CODE holds one-line App_* hooks only, so regeneration cannot eat logic); app_config.h (pins, ADC map, Rev 1 quirks, backstop limits, PWM lock)
Core	app_main.c (boot, scheduler, IWDG, LEDs, Error_Handler policy); app_control.c (state machine, ARM/RUN rules, control ISR, modes, plant sim); app_focmath.h (Clarke/Park, SVM, PI, PLL, dq model; host-tested); app_pwm.c (sole output-enable path)
Safety	app_safety.c (fault bits, invariant, protections, ISR→main fault hand-off); app_selftest.c (reset cause, UID, option bytes, dead-time and break read-back)
Drivers	app_analog.c (ADC1/2, CSA cal); app_as5047p.c (SPI1, pipelined read, PLL velocity, multi-turn); app_drv8323.c (SPI3, bounded, verify, pin walk); app_uart.c (USART1 + DMA2, register level); app_can.c (FDCAN1, FOCCAN/1, run-time bit timing); app_time.c (64-bit DWT)
Data	app_signals (named globals); app_registry + registry_gen.c; app_params (flash page 62 at 0x0801F000, by ID, CRC); app_foclink; app_scope (stream + triggered capture)

13.2 Timing

```
TIM1 centre-aligned 25 kHz -> CH4 at peak -> TRGO2 -> ADC injected -> ADC1 JEOS IRQ (prio 0) -> Ctrl_Isr
Main loop: FOCLINK, invariants, state machine, CAN every pass; slow ADC + protections every 10 ms
Motor-ID step logic runs in the main loop behind a handshake; the ISR keeps only per-tick work
```

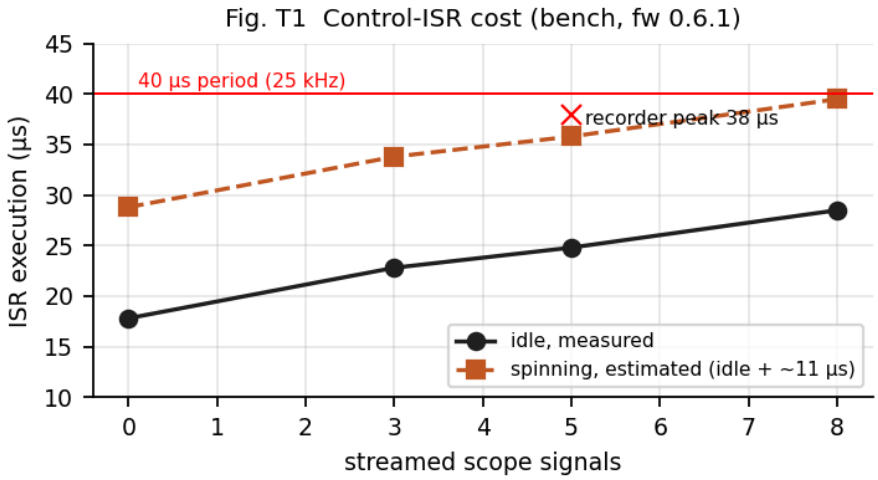


Fig. T1. Control-ISR cost vs streamed scope signals (measured idle; spinning estimated as idle + ~11 µs). The 8-signal recorder during a spin exceeded the 40 µs period and tripped FLT_ISR.

- ISR worst case cut from ~27 to ~15 µs: removed newlib fmodf; polynomial fm_sincos (~700 → ~270 cycles); force-inline helpers (at -Og static inline became veneered flash calls); hot path in SRAM; flash prefetch on.
- Rule: add nothing to the ISR without measuring isr_peak_us.

13.3 States and modes

```
BOOT -> DISARMED --ARM--> ARMED --RUN mode--> RUNNING
RUNNING --heartbeat lost--> DISARMED ;  any --fault--> FAULT --clear--> DISARMED
```

Mode	Function	Status
PWM_TEST	Fixed duty for p_pwm_test_ms	Approved, validated
PHASE_CHECK	Sign/wiring check along each phase vector	Approved, validated (error 1–2 %)
ID_R / ID_L	Regulate only the in-axis current, perpendicular axis at 0 V; accept a hold only if voltage agrees across both halves of the window	Approved, validated
IF_SPIN	Open-loop I/f, delay-compensated, tiered caps	Approved, to 2000 rpm at 12 V
CSA_CAL / ALIGN	Calibration, rotor alignment	Defined
TORQUE / VELOCITY / POSITION	Closed-loop FOC	Not run

13.4 Data model and interfaces

Item	Detail
Signal registry	docs/signal_registry.csv → tools/gen_registry.py → registry_gen.c. A row without a same-named, same-typed C global fails the build.
FOCLINK/1	CRC-8 text lines + binary SCOPE/CAPTURE frames. @INFO (registry CRC32), @SIG, @C, @P, @S (500 ms), @D, @EV, @ACK with reason code. Commands ARM, RUN, REF, SET, TUNE, SCOPE, CAPTURE, DRV, ENC, CAN, SELFTEST, FAULTLOG, REBOOT. Heartbeat required while ARMED.

Item	Detail
FOCCAN/1	1 Mbit/s, 80 % sample point default; bit timing from the real kernel clock; node ID 63 = unassigned; loopback self-test.
Parameters	Stored by ID in their own flash page; firmware image never overlaps it. SAVE only when DISARMED (stalls ISR ~25 overruns); read back RAM before SAVE; back up page before flashing.
Plant simulation	p_sim_enable = 1 substitutes a dq motor model; validated motor ID to R within 0.15 %, L within 0.4 %.
Build / resources	Flash ≈ 107 of 124 KB, RAM ≈ 98 % (bench, -Og). -O0 does not link. Off-IDE ARM build (tests/arm_build) and STM32_Programmer_CLI flash/readback. No host C compiler on the dev PC: tests/host not run.

14 PC GUI (FOCLINK)

- Python, PySide6 + pyqtgraph; Windows and macOS; packaged as a desktop app.
- Layers: transport (background thread, auto-baud 921600/115200 with traffic scoring, heartbeat) → protocol decoder → store (stale-link detection, missed-frame count, CSV frame log at 2 Hz) → view model (valid-when rule per value; invalid shows a dash with the reason) → widgets.
- Pages: live dashboard; guided motor bring-up (phase check, R, L); spin control "cockpit" throttle deck that computes and stops below the voltage-limited speed; 500 Hz recorder with back-EMF analysis (range limit 1250 rpm).
- Single engineer view; NERV-style ops-panel theme with EN/JP labels and status animations.
- Firmware simulator in Python for board-less testing; 85 automated tests including real mouse clicks. Tests cross-check simulator limits against app_config.h and its reason table against app_control.c.
- Lessons: frame logs found the frozen-ADC, voltage-limited trips and recorder overload after the fact. Look at what does not change. Timing-sensitive tests must wait for conditions, not read-and-assert.

15 PCB and Fabrication

Item	Value
Size / layers	70.10 × 70.10 mm, 3.2 mm corner radius, 6 layers
Stackup	Sig/Pwr · GND · Sig/Pwr · Sig/Pwr · GND · Sig/Pwr
Copper	2 oz outer assumed by all thermal/current figures (order parameter; confirm)
Vias	344 × 0.25 mm drill / 0.55 mm pad (0.15 mm annular, at JLCPCB minimum)
Via-in-pad	16 per high-side drain inside one 4.41 × 3.73 mm mask opening; needs resin-plugged, capped vias or solder wicks
Stencil	Windowpaned on every thermal pad (FETs, DRV8323, STM32)
Placement	159 top / 33 bottom; AS5047P at (40.00, 40.00), bottom
Power terminals	Phases and V_{BUS} on one edge (y = 71 mm), adjacent to FET row
Commutation loop	MLCCs at y 56.25, ~6 mm from FETs; estimated ~2.4 nH
Chassis	R58 0 Ω at H6 only; C73–C76 1 nF / 1 kV; R56/R57/R59 DNP; moat cut on inner planes
Test points	38
DNP	Snubber RC (R23/24/28/29/33/34, C35/36/45/46/56/57), J2, J7, R56/57/59
Assembly	JLCPCB PCBA; ISC030N10NM6 consigned/extended

15.1 BoM consolidation practice

- Done once the design was stable. Merge duplicate rows; parallel/series synthesis from existing values (e.g. 49.9 k from 2 × 100 k, 0.2 % shift); footprint unification; X5R → X7R where hot.
- Declined where the reason for the part governs: trip-point dividers, USB-C Rd, gate resistors, series high-voltage divider legs (0402 working-voltage limit).
- Divider changes alter firmware scaling constants; the V_{BUS} lower leg moved 2.2 k → 2 k.

16 Bring-Up and Test Results

16.1 Timeline

Date	Firmware	Milestone
before 09-22	—	Plan: self-test → phase check → R/L → open loop → λ → closed loop. I/f chosen over V/f.
09-23	0.4.0	Modular architecture, registry, FOCLINK/1, params, plant sim, CAN, scope, DWT time base
09-23	—	Board talks: wrong DRV_ENABLE/encoder-CS pins fixed; SPI3 dropped to 332 kHz; ISR overruns ~17 000/s → 0
09-23	0.5.0	First switching: single enable path; JSQR.JL fix; ARM waits for nFAULT; self-measured dead time 200 ns, zero overlap
09-26/27	0.6.0	Motor ID: normal-mode CSA cal; sign fix; R, L measured; Ctrl_Arm race fixed
09-27/28	—	IF_SPIN tiers 1–3; delay compensation; 2000 rpm
09-28	0.6.1	Voltage-limited trips explained; frozen slow-ADC readings fixed; staleness supervision

16.2 Spin tiers

Tier	Limits	Result
1	60 rpm, 0.5 A, 1 V, 6 s	No motion: 0.5 A ≈ 13 mN·m vs ~37 mN·m breakaway
1b	2 A cap, staged	Spun at ≤ 1.5 A; 5 turns in 5 s at 60 rpm confirms 14 pole pairs
2	300 rpm, 3 A, 12 s	FLT_ISR with 8 scope signals; clean with 5. λ = 1.389 mV·s measured
3	2700 rpm cap, 6 V, 300 rpm/s	Hunting ~19 Hz above ~300 rpm, pull-out 530–700 rpm; fixed by delay compensation (recorded to 1200 rpm)
bench	p_v_limit from flash = 2 V	Three OC trips at 955–980 rpm (voltage-limited pull-out); 2000 rpm with limit raised in RAM

16.3 Bench motor characterisation (T-Motor MN601-S KV320)

Parameter	Datasheet	Measured	Method
Pole pairs	14	14	Turn count at 60 rpm
R per phase	39 mΩ	98 mΩ	ID_R ×2; DC test 0.18/0.37/0.57 V at 1/2/3 A (195 mΩ L-L)
L_d / L_q	—	7.9 / 11.0 µH	ID_L ×2 (firmware uses 9.44 µH)
λ	—	1.389–1.414 mV·s	Back-EMF, 500 Hz recorder
K_V / K_t	320 rpm/V / —	280–284 rpm/V / 0.0292 N·m/A	From λ; $K_t = 1.5 \cdot p \cdot \lambda$
Inertia / breakaway	—	≈ 8e-5 kg·m² / ≈ 37 mN·m	I/f swing / load angle

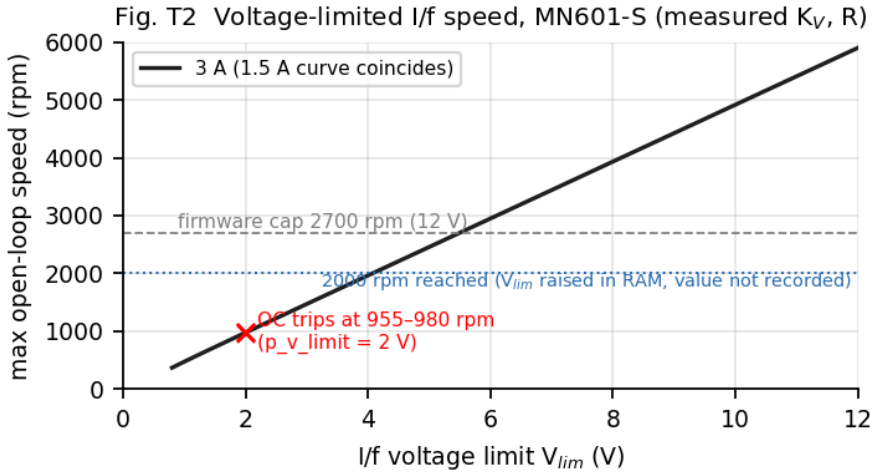


Fig. T2. Voltage-limited open-loop speed with measured K_V and R. The three trips at 955–980 rpm sit on the 2 V curve.

16.4 Findings worth carrying forward

- Trust the schematic over the code pin map: two wrong pins made a healthy driver look dead.
- Open-drain SDO + pull-up sets the SPI ceiling; one-bit-shifted reads are the signature.
- Do not trust HAL to program what was asked: read registers back and check at boot (JSQR.JL = 0).
- Polling a multi-conversion ADC sequence from a preemptible context loses EOCs silently; every "fail → keep old value" path needs a counter and an age.
- ADC_CR has set-only bits; read-modify-write re-issues them. Mask as the LL driver does.
- Protections are only as good as their inputs: OV/UV/OT read a frozen number for five days.
- Measure before streaming: each scope signal costs ~1 µs of a 40 µs ISR.
- Datasheet constants were off (K_V 11 %, R 2.5×): use measured values for limits; cross-check with a meter.
- Own mistakes: misread supply current by 10×; saved stale parameters after an unnoticed reset; left p_v_limit = 2 V in flash.

17 Known Issues (Errata)

ID	Issue	Impact	Workaround / Rev 2
E-01	TVS clamps 82.4 V; DRV8323 and LMR36520 abs max 65 V	No protection 65–82 V	Short leads, no hot-plug; revise clamp
E-02	PC3/BKIN2 E-stop follows DRV_ENABLE via R41	No independent E-stop	Supply switch; Rev 2 real input
E-03	Low-side FETs have no thermal vias	Limits continuous current	Rate 10 A; add vias / bottom pour
E-04	DRV8323 SDO open-drain, R17 too slow at 2.66 MHz; no DRV_CS pull-up	Shifted reads	SPI3 332 kHz; resize R17, add pull-up
E-05	PB8 FDCAN_RX shared with BOOT0	Powered CAN can boot ROM loader	nSWBOOT0 = 0, nBOOT0 = 1
E-06	DRV8323 CAL-mode offsets wrong	0.15–0.46 A error	Normal-mode cal; cleared every ARM

ID	Issue	Impact	Workaround / Rev 2
E-07	Phase C CSA reversed; A/B datasheet signs inverted	Wrong current loop	Signs +1/+1/−1
E-08	Analog shares digital 3V3	Reference ripple on non-ratiometric channels	Separate analog rail
E-09	Bulk 3 × 100 µF vs 5 specified	~96 % ripple loading	Measure; five pieces
E-10	Board NTC at board centre	Under-reads FETs	Calibrate offset; move NTC
E-11	No top fiducials; no silkscreen designators	Assembly/rework	Rev 2
F-01	PA9/PA10 high enable UCPD pulldowns on PB6/PB4	UART TX / LED pulled low	Set UCPD1_DBDIS before TIM1
F-02	HAL left ADC1 JSQR.JL = 0	Phase C, fast V_{BUS} read 0	Write JSQR; check at boot
F-03	Slow ADC sequence polled under preemption	Readings frozen 5 days	One conversion per start; staleness
F-04	ADC_CR set-only bits in read-modify-write	Spurious starts	Mask rs bits
F-05	Flash SAVE stalls ISR	~25 overruns	SAVE when DISARMED, power-cycle
F-06	HAL uwTick seen frozen	HAL timing unreliable	DWT time base

18 Open Items and Rev 2

18.1 Open

- Switch node never scoped (DMM read up to 75 V on a 24 V bus, likely artefact). Scope before any load or higher voltage, with a differential GHx–SHx probe at a low-side turn-on edge.
- Intermittent PWM_ENABLE refusal after boot (2 of 20); board reset while armed (12 W supply suspected).
- L_d/L_q ratio 1.4 on a surface-magnet outrunner; datasheet R 39 mΩ vs 98 mΩ.
- Encoder magnet, closed-loop FOC, CAN on a bus, locked-rotor torque test (proposed, not approved).
- Firmware could clamp I/f speed to the voltage-feasible speed itself (a reviewed change).
- Next measurements: low-side case temperature at 10 A, R37 offset, encoder angle error vs I_q, bulk ripple current.

18.2 Rev 2 changes

- Independent E-stop input with ESD/RC
- Low-side thermal vias or bottom switch-node pour; NTC in FET zone
- R17 sized for faster SPI; DRV_CS pull-up
- Separate analog 3V3; CSA filter/bias review
- BOOT0 not shared with CAN RX
- Five bulk capacitors, more bus capacitance or a brake chopper
- Input clamp below 65 V
- Hardware CAN node ID
- Fiducials, silkscreen designators, 0.3/0.6 mm vias

19 Corrections to Earlier Project Documents

Topic	Earlier statement	Corrected
Switching frequency	40 kHz PWM, 20 kHz loop	25 kHz PWM and control ISR
Q_g ceiling	~52 nC at 2× ($6 \cdot Q_g \cdot f$)	Charge pump drives 3 high-side gates: ~104 nC at 40 kHz, ~167 nC at 25 kHz
Shoot-through	0.06 V headroom, raise IDRIVE to 1.0/2.0 A	Driver-model artefact; keep 0.33/0.66 A until scoped (Fig. S2, S3)
Snubber footprints	Gate-source RC	Drain-source RC (bus–phase, phase–source)
Current range	±27.5 A	±27.5 A ideal, ±23.3 A linear
Continuous bus limit	56 V	51 V (TVS stand-off)
Buck rating	80 V part	LMR36520, 65 V abs max
E-stop	Conditioned input, check polarity	Non-functional on Rev 1
DRV SPI zeros	Open issue	Resolved: wrong pin map + slow pull-up
Thermal rating	20 A comfortable	10 A until low-side measured
GUI tests	19	85

20 Revision History and References

Version	Date	Change
Datasheet 1.0	2026-09-29	IC-style datasheet
TRM 1.0	2026-09-30	Consolidated reference; ngspice re-simulation; corrections table

Devices: TI SLVSDJ3D (DRV8323), SLVA714 (IDRIVE/TDRIVE), SLVAF66 (motor drive design); Infineon ISC030N10NM6 Rev 2.2; ST DS12589 (STM32G431), RM0440, AN2867 (oscillators), AN4013 (timers), AN2834 (ADC accuracy); ams AS5047P DS000324; TI TCAN1044A, LMR36520, TLV757P. **Theory:** TI BPRA048 (Clarke/Park); SimpleFOC documentation; H. Ott, *Electromagnetic Compatibility Engineering*; ADI MT-031 (grounding). **Standards:** ISO 11898 (CAN), IEC 60204-1 (stop categories), JESD51-1 (thermal), IPC-2221. **Project:** DDR-001, UG-001, Rev 1 Layout Review, docs/FIRMWARE.md, docs/firmware_context.md, CLAUDE.md, simulation/ folder.